

ACE Services Portal

Final Feature Verification Checklist

Construction Estimation & CAD Design Portal

STATUS Development Complete — Ready for Deployment

DATE August 25, 2026

PREPARED FOR Executive Review & Sign-Off

VERSION 1.0 Final

Executive Summary

The ACE Services Portal is a **complete, production-ready SaaS application** for managing construction estimation and CAD design projects from client intake through delivery. The system handles multi-role workflows with secure authentication, comprehensive audit logging, financial tracking, and automated email notifications.

Core Value Propositions

- ✓ End-to-end project lifecycle management (from intake to client delivery)
 - ✓ Role-based access control preventing unauthorized data access
 - ✓ Automated notifications and email delivery tracking
 - ✓ Financial management with merchant fee tracking
 - ✓ Comprehensive audit trail for compliance
 - ✓ Professional reporting and KPIs
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1. User Management & Authentication

1.1 User Roles & Access Control

- ✓ **BD Agent** — Business development; can submit projects, upload intake files, track their projects
- ✓ **Estimation Engineer** — Can view assigned projects, upload deliverables, raise RFIs; blocked from seeing client details/pricing
- ✓ **Design Engineer** — Same as estimation engineer; supports CAD design & drafting projects
- ✓ **Admin** — Full system access; assign projects, manage users, set pricing, send deliverables, view audit logs
- ✓ **Privacy Enforcement**: Engineers never see client email, phone, company name, salesperson, or pricing — enforced at the service layer via `toRoleView()`.

1.2 Authentication System

- ✓ **Login** — Email + password authentication
- ✓ **Account Lockout** — After 5 failed login attempts, account locked for 15 minutes + email notification sent
- ✓ **JWT Tokens** — Short-lived access tokens (15 minutes) + 7-day refresh tokens
- ✓ **Session Management** — Refresh tokens revoked on logout; all tokens revoked on password reset
- ✓ **HTTP-Only Cookies** — Refresh tokens stored as secure HTTP-only cookies
- ✓ **Rate Limiting** — Auth endpoints throttled (20 requests per 15 minutes) to prevent brute force attacks

1.3 Password Management

- ✓ **Password Reset** — Email-based flow with 60-minute expiry on reset link
- ✓ **Password Reset Tokens** — Hashed in database; marked `used=true` after redemption
- ✓ **Setup Workflow** — Admins create users, personalized setup email sent, employee sets own password to activate

1.4 Account Lifecycle

- ✓ **User Creation** — Admin creates user with email + role; pendingSetup=true
- ✓ **Setup Email** — Personalized activation link sent to user
- ✓ **Account Activation** — User sets password, account becomes active
- ✓ **Deactivation** — Soft deactivate (preserves history, prevents login)
- ✓ **Hard Delete** — Removes user; reassigns BD projects to admin, nullifies history references

1.5 Audit Logging

- ✓ **Login Success** — Logged with user ID and role
- ✓ **Login Failure** — Logged with failure reason (wrong password, user inactive) and fail count
- ✓ **Password Reset Request** — Logged with user ID
- ✓ **Password Reset Completion** — Logged with user ID
- ✓ **Account Lockout Email** — Sent when lockout triggered
- ✓ **All User Modifications** — Account creation, deactivation, deletion, role changes logged with actor and timestamp

2. Project Management

2.1 Project Creation & Intake

- ✓ **Submit Project** — BD agents submit projects with client details (name, email, phone, company), scope, requested deadline, price, salesperson
- ✓ **Reference Number** — Auto-generated with year-based sequence (PRJ-2025-0001); globally unique
- ✓ **Project Type** — Two categories: "Cost Estimation" or "CAD Design & Drafting"
- ✓ **Intake Files** — BD agents upload client drawings, specifications, RFQ documents
- ✓ **File Categories** — Supports: drawing, specification, takeoff, estimation_sheet, client_doc
- ✓ **Presigned S3 URLs** — Secure upload/download mechanism with 72-hour expiry for deliverables
- ✓ **Demo Mode** — Local filesystem upload/download for development (no S3 required)

2.2 Project Workflow & Status Transitions

- ✓ **Status States** — received → proposal → follow_up → approved → declined → assigned → in_progress → delivered → sent_to_client
- ✓ **BD Sales Cycle** — BD agent can update project status during sales phase (follow-up dates, notes)
- ✓ **Admin Assignment** — Admin assigns project to appropriate engineer (estimation or design)
- ✓ **Set Deadlines** — Admin sets internal deadline (different from client-requested deadline)
- ✓ **Priority Levels** — low, medium, high, urgent; visible to assigned engineer
- ✓ **Admin Instructions** — Free-form text instructions; visible to engineer but not to other roles
- ✓ **Status History** — Complete audit trail of all status transitions with actor, timestamp, notes

2.3 Engineer Assignment & Work Flow

- ✓ **Receive Assignment** — Engineer receives in-app + email notification when project assigned
- ✓ **View Deadline Countdown** — Engineers see days and hours remaining to internal deadline
- ✓ **Mark In Progress** — Engineer marks project as in_progress when starting work
- ✓ **Upload Deliverables** — Engineer uploads final files (estimates, CAD files, etc.)
- ✓ **Mark Delivered** — Engineer marks project delivered (requires at least one deliverable file)
- ✓ **Reassignment** — Admin can reassign project to different engineer (tracked in history)

2.4 Project Visibility by Role

- ✓ **BD Agents** — See only their own projects; full details (client, pricing, salesperson, status)
- ✓ **Engineers** — See only their assigned projects; blocked from client details/pricing/salesperson
- ✓ **Admin** — See all projects; full details including audit trail
- ✓ **Role-Based View Filtering** — Enforced at service layer; cannot be bypassed via API

2.5 Project Search & Filtering

- ✓ **Search** — By reference number, client company, client contact person, salesperson
- ✓ **Filter by Status** — received, proposal, follow_up, approved, declined, assigned, in_progress, delivered, sent_to_client
- ✓ **Filter by Project Type** — Cost Estimation vs. CAD Design & Drafting
- ✓ **Filter by Date Range** — submitted_at between start and end dates
- ✓ **Pagination** — 100 projects per page (configurable up to 200)

2.6 Project Deletion

- ✓ **Hard Delete** — Admin can delete project; cascades to status history, files, deliverables, RFIs
- ✓ **Audit Logged** — Deletion event logged with reference number

3. Request for Information (RFI) System

3.1 RFI Creation & Management

- ✓ **Create RFI** — Engineer can raise RFI with title, question, optional attachment
- ✓ **Attachment Support** — RFI attachment with metadata (name, S3 key, MIME type, upload time)
- ✓ **Admin Notification** — Admin notified immediately (in-app + email) when RFI created
- ✓ **Status Tracking** — RFI states: pending → answered OR forwarded_to_client

3.2 Admin RFI Response

- ✓ **Direct Answer** — Admin can provide answer directly in portal (status: answered)
- ✓ **Engineer Notification** — Engineer notified when RFI answered

- ✓ **Forward to Client** — Admin can forward question to client for clarification
- ✓ **Client Forwarding** — External email sent to client with RFI question + optional attachment
- ✓ **RFI History** — Complete audit trail of all RFI activity per project

3.3 RFI Audit Logging

- ✓ **RFI Created** — Logged with engineer, project, title
- ✓ **RFI Answered** — Logged with admin answer text
- ✓ **RFI Forwarded** — Logged when forwarded to client

4. Deliverable Management & Client Delivery

4.1 Deliverable Upload

- ✓ **Engineer Uploads** — Assigned engineer uploads final deliverable files
- ✓ **File Metadata** — Stores original name, S3 key, MIME type, file size, upload timestamp
- ✓ **Multiple Files** — Project can have multiple deliverables (all sent together to client)

4.2 Delivery Preview & Preparation

- ✓ **Preview Screen** — Admin previews all deliverables + auto-generated invoice before sending
- ✓ **Invoice Generation** — Auto-generated based on base project fee, merchant/processing fee, and total due
- ✓ **Invoice Customization** — Admin can edit invoice text or provide custom invoice
- ✓ **Scope Display** — Auto-populated from project description

4.3 Merchant Fee Management

- ✓ **Fee Type** — Support both percentage-based and fixed-amount fees
- ✓ **Per-Project Fees** — Each project can have different fee structure
- ✓ **Total Calculation** — Total due = base price + merchant fee amount
- ✓ **Audit Trail** — Merchant fee changes tracked in project record

4.4 Flexible Delivery Methods

- ✓ **Email Attachment** — If total deliverable size < 25 MB, send files as email attachments
- ✓ **Download Links** — If total > 25 MB, send 72-hour pre-signed S3 download links
- ✓ **Automatic Selection** — System recommends method based on file size; admin can override
- ✓ **Custom Email Subject & Body** — Admin can customize email message before sending

4.5 Client Delivery Execution

- ✓ **Send to Client** — Admin clicks "Send to Client"; email sent to client_email address

- ✓ **Delivery Log** — Every delivery tracked: recipient email, subject, delivery method, success/failure, error message
- ✓ **Status Transition** — Project status automatically updated to sent_to_client on successful delivery
- ✓ **Success Verification** — API returns success/failure with error details if email failed

4.6 Email Delivery Tracking

- ✓ **Resend Webhook** — Receive webhook events from Resend (delivery, bounce, opened, etc.)
- ✓ **Event Logging** — Email events logged to audit trail with recipient, event type, timestamp
- ✓ **In-App Notifications** — Email events appear in admin notification center

5. Financial Management & Pricing

5.1 Project Pricing

- ✓ **Base Price** — Decided price per project; stored as decimal with 2 decimal places
- ✓ **Merchant Fee** — Percentage-based fee (stored in merchantFeePercent) OR fixed amount
- ✓ **Fee Calculation** — Fee amount = base price × (merchantFeePercent / 100) OR fixed amount
- ✓ **Total Due** — Calculated as base price + merchant fee amount
- ✓ **Price Visibility** — Visible to BD agents and admins; hidden from engineers

5.2 Invoice Generation

- ✓ **Auto-Generated Invoice** — System generates professional invoice with client info, reference, service type, scope, fees and total due
- ✓ **Custom Invoice** — Admin can provide custom invoice text (replaces auto-generated)
- ✓ **Invoice Storage** — Saved in project record; can be updated before sending to client

5.3 Financial Reporting

- ✓ **Weekly Excel Export** — Admin can export comprehensive weekly reports: all projects, KPI summary, department breakdown, salesperson breakdown
- ✓ **Time Range Options** — this_week, last_7_days, last_week, this_month, all_time, custom date range
- ✓ **Excel Formatting** — Professional headers, color-coded summary, totals row

6. Notifications & Communication

6.1 Email Service

- ✓ **Multi-Provider Support** — Fallback chain: Resend → SendGrid → SMTP → Demo console
- ✓ **Email Features** — Text, HTML, file attachments, custom headers
- ✓ **Environment Configuration** — All providers configured via .env variables
- ✓ **Demo Mode** — Emails logged to console for development (no external email provider required)

6.2 Automated Email Notifications (11 triggers)

Trigger	Recipient	Subject Line
New project submitted	Admin	New Project Uploaded: {ref} — {company}
Project assigned to engineer	Assigned Engineer	New Assignment: {ref} ({service_type})
RFI created by engineer	Admin	New Engineering RFI: {ref} — {title}
RFI answered by admin	Engineer who raised RFI	RFI Answered: {ref}
RFI forwarded to client	Client	Inquiry / Request for Information — Project {ref}
Deliverables ready for dispatch	Admin	In-app notification (email optional)
Deliverables sent to client	Client	Your Project {ref} Deliverables & Invoice — {company}
Account locked (5 failed logins)	User	Account Locked
Password reset requested	User	Password Reset Request
New user account created	New User	Welcome to ACE SERVICES — Activate Your {role} Workspace
Email delivery events	Admin	Email {event_type}

6.3 In-App Notifications

- ✓ **Notification Center** — Users can view all notifications in portal
- ✓ **Read Status** — Notifications can be marked as read individually or all at once
- ✓ **Event Types** — PROJECT_SUBMITTED, PROJECT_ASSIGNED, RFI_CREATED, RFI_ANSWERED, PROJECT_DELIVERED, EMAIL_EVENT, etc.
- ✓ **Metadata** — Notifications include links and context (projectId, referenceNumber, etc.)
- ✓ **Pagination** — Notifications fetched with pagination

7. Audit & Compliance

7.1 Comprehensive Audit Logging — 18 Event Types

USER_LOGIN_SUCCESS	USER_LOGIN_FAILURE	PASSWORD_RESET_REQUEST
PASSWORD_RESET_COMPLETE	PROJECT_SUBMITTED	PROJECT_STATUS_UPDATED
PROJECT_ASSIGNED	PROJECT_REASSIGNED	DELIVERABLE_UPLOADED
SEND_TO_CLIENT_SUCCESS	SEND_TO_CLIENT_FAILURE	RFI_CREATED
RFI_ANSWERED	RFI_FORWARDED	USER_ACCOUNT_CREATED
USER_ACCOUNT_UPDATED	USER_ACCOUNT_DEACTIVATED	ROLE_CHANGED
USER_ACCOUNT_DELETED		

7.2 Audit Log Data

- ✓ **Actor Identification** — Event logs include actor ID (user who performed action) and actor role
- ✓ **Timestamp** — All events timestamped (UTC with timezone info)
- ✓ **Target Identification** — Target ID (project, user, etc.) tracked
- ✓ **Metadata** — JSON metadata for rich context (email addresses, failure reasons, fee changes, etc.)

7.3 Audit Log Queries

- ✓ **Audit Endpoint** — /audit endpoint for filtering audit logs
- ✓ **Filter by Event Type** — Search specific event types
- ✓ **Filter by Actor** — Find all actions by specific user
- ✓ **Filter by Date Range** — Events between start and end dates
- ✓ **Pagination** — Audit logs paginated for performance
- ✓ **Admin-Only Access** — Only admins can view audit logs

8. File Management

8.1 File Upload Infrastructure

- ✓ **Dual storage backends**: Production (AWS S3 / Cloudflare R2) and Development (local filesystem — demo-uploads/)
- ✓ **Presigned URLs** — Secure upload mechanism without exposing credentials
- ✓ **MIME Type Validation** — Files stored with MIME type
- ✓ **File Size Tracking** — Each file records size in bytes

8.2 Upload Workflow

- ✓ **Step 1: Request Upload URL** — POST /files/upload-url with project ID, file name, MIME type → receive presigned URL
- ✓ **Step 2: Upload File** — Client uploads file via PUT to presigned URL
- ✓ **Step 3: Confirm Upload** — POST /files/confirm with file metadata → stored in database

8.3 Download Workflow

- ✓ **Request Download URL** — GET /files/:id/download-url → receive presigned URL (72-hour expiry)
- ✓ **Security** — Files can only be downloaded by authorized users (role-based)
- ✓ **Audit Trail** — Download requests can be tracked (optional implementation)

8.4 File Categorization

- ✓ **Intake files** (uploaded by BD agent): drawing, specification, takeoff, estimation_sheet, client_doc
- ✓ **Deliverables** (uploaded by engineer): final files, same categorization available

8.5 File Deletion

- ✓ **Admin Deletion** — Admin can delete files from project
- ✓ **Cascade Deletion** — When project deleted, all associated files cascade deleted
- ✓ **S3 Cleanup** — Files removed from S3 storage (or demo-uploads)

8.6 Demo Mode

- ✓ **Local Upload** — PUT /files/demo-upload for development
- ✓ **Local Download** — GET /files/demo-download for development
- ✓ **No S3 Required** — Development can proceed without AWS credentials

9. Security & Compliance

9.1 Authentication Security

- ✓ **Password Hashing** — bcrypt with 12 salt rounds (industry standard)
- ✓ **Token Hashing** — Refresh tokens hashed before storage (prevents session fixation)
- ✓ **JWT Signing** — Access tokens signed with JWT_SECRET
- ✓ **Token Expiry** — Access tokens expire in 15 minutes; refresh tokens in 7 days
- ✓ **HTTPS Enforcement** — Secure flag on HTTP-only cookies (production)
- ✓ **SameSite Policy** — Cookies set to Lax (CSRF protection)

9.2 Authorization & Access Control

- ✓ **Role-Based Access Control (RBAC)** — 4 user roles with distinct permissions
- ✓ **JWT Guards** — All protected endpoints require valid JWT access token
- ✓ **Role Guards** — Endpoints restricted by role (e.g., only admins can assign projects)
- ✓ **Ownership Verification** — Engineers can only see their own assigned projects
- ✓ **Cascading Permissions** — Admins have superset of all permissions

9.3 Input Validation

- ✓ **DTO Validation** — Class validators on all request DTOs
- ✓ **Email Format Validation** — Email fields validated with regex
- ✓ **Enum Validation** — Role, status, priority validated against enums
- ✓ **String Length Limits** — VarChar constraints prevent oversized inputs

9.4 SQL Injection Prevention

- ✓ **Parameterized Queries** — Prisma ORM prevents SQL injection via prepared statements
- ✓ **No Raw SQL** — Raw queries use parameterized values (\$1, \$2, etc.)
- ✓ **Input Sanitization** — All user input sanitized before database queries

9.5 Rate Limiting

- ✓ **Global Rate Limit** — 5000 requests per minute per IP
- ✓ **Auth Throttling** — Auth endpoints throttled to 20 requests per 15 minutes
- ✓ **Prevents Brute Force** — Account lockout after 5 failed login attempts

9.6 Error Handling

- ✓ **Generic Error Messages** — Login failures don't reveal which field is wrong
- ✓ **No Password Resets on Inactive Users** — Prevents email enumeration
- ✓ **Secure Error Logs** — Detailed errors logged server-side; generic errors returned to client

9.7 Data Protection

- ✓ **Soft Deletes** — Users soft-deleted by default (data preserved for audit)
- ✓ **Confidential Fields** — Engineer views explicitly exclude client email, phone, pricing
- ✓ **No Sensitive Data in URLs** — Tokens passed in headers/cookies, not URL parameters
- ✓ **CORS** — Pre-flight support configured for frontend communication

10. API Endpoints Checklist

10.1 Authentication

✓ POST /auth/login — Email + password login
✓ POST /auth/refresh — Refresh access token
✓ POST /auth/logout — Revoke refresh token
✓ POST /auth/forgot-password — Send password reset email
✓ POST /auth/reset-password — Set new password with reset token
✓ POST /auth/check-email — Check if email is pending setup
✓ POST /auth/complete-setup — Activate account, set password

10.2 Project Management

✓ POST /projects — Create project (BD Agent, Admin)
✓ GET /projects — List projects (filtered by role)
✓ GET /projects/:id — Get project details (role-filtered view)
✓ PATCH /projects/:id/status — Update project status (BD Agent)
✓ PATCH /projects/:id/merchant-fee — Set merchant fee (Admin)
✓ PATCH /projects/:id/assign — Assign project to engineer (Admin)

- ✓ PATCH /projects/:id/reassign — Reassign to different engineer (Admin)
- ✓ PATCH /projects/:id/mark-in-progress — Mark as in_progress (Engineer)
- ✓ PATCH /projects/:id/mark-delivered — Mark as delivered (Engineer)
- ✓ DELETE /projects/:id/files/:fileId — Delete intake file (Admin)
- ✓ GET /projects/:id/status-history — View status transitions
- ✓ GET /projects/export/weekly-excel — Export Excel report (Admin)

10.3 RFI

- ✓ POST /projects/:id/rfis — Create RFI (Engineer)
- ✓ GET /projects/:id/rfis — List RFIs (all authenticated users)
- ✓ PATCH /projects/:id/rfis/:rfid/answer — Admin answers RFI (Admin)
- ✓ POST /projects/:id/rfis/:rfid/forward-client — Forward to client (Admin)

10.4 Delivery

- ✓ GET /delivery/:projectId/preview — Preview invoice + deliverables (Admin)
- ✓ POST /delivery/:projectId/send — Send to client (Admin)
- ✓ POST /delivery/resend-webhook — Receive email events (public webhook)

10.5 Files

- ✓ POST /files/upload-url — Request presigned S3 upload URL
- ✓ POST /files/confirm — Confirm upload, store metadata
- ✓ GET /files/:id/download-url — Request presigned download URL
- ✓ DELETE /files/:id — Delete file (Admin)
- ✓ PUT /files/demo-upload — Demo-mode local file upload
- ✓ GET /files/demo-download — Demo-mode local file download

10.6 User Management

- ✓ GET /users — List all users (Admin)
- ✓ GET /users/engineers — List active engineers (Admin)
- ✓ POST /users — Create user (Admin) → triggers setup email
- ✓ PATCH /users/:id — Update user name/role (Admin)
- ✓ DELETE /users/:id/deactivate — Soft deactivate (Admin)
- ✓ DELETE /users/:id — Hard delete (Admin)

- ✓ POST /users/:id/reset-password — Trigger password reset email (Admin)
- ✓ PATCH /users/profile — Update own name/password (authenticated user)

10.7 Notifications

- ✓ GET /notifications — Fetch user's notifications (pagination)
- ✓ PATCH /notifications/:id/read — Mark notification as read
- ✓ PATCH /notifications/mark-all-read — Mark all as read

10.8 Audit

- ✓ GET /audit — Query audit log with filters (Admin only)

11. Database Schema

11.1 Tables (11)

- ✓ users — Full name, email (unique), role, active status, pending setup, lockout tracking
- ✓ refresh_tokens — Hashed tokens, expiry, revoke flag
- ✓ password_reset_tokens — Hashed tokens, expiry, used flag
- ✓ projects — Reference number, client info, pricing, scope, deadlines, status, type
- ✓ project_files — Intake drawings; S3 key, MIME type, size, category
- ✓ deliverables — Engineer-uploaded final files; same metadata as project files
- ✓ project_status_history — Status transitions with actor, timestamp, notes
- ✓ project_rfis — RFI details; title, question, attachments, status, answers
- ✓ client_delivery_log — Email delivery records; recipient, method, success, error
- ✓ notifications — In-app notifications; event type, title, body, read status
- ✓ audit_log — Audit trail; event type, actor, timestamp, metadata

11.2 Enums

- ✓ UserRole — BD_AGENT, ESTIMATION_ENGINEER, DESIGN_ENGINEER, ADMIN
- ✓ ProjectStatus — received, proposal, follow_up, approved, declined, assigned, in_progress, delivered, sent_to_client
- ✓ PriorityLevel — low, medium, high, urgent
- ✓ AuditEventType — 18+ event types (see Section 7.1)

11.3 Relationships

- ✓ Users → one-to-many: projects submitted, projects assigned, status changes, deliverables, notifications, audit entries
- ✓ Projects → one-to-many: status history, files, deliverables, RFIs, client delivery logs

- ✓ Refresh Tokens → many-to-one: user
- ✓ Password Reset Tokens → many-to-one: user
- ✓ All foreign keys configured with CASCADE delete where appropriate

11.4 Indexes

- ✓ Performance indexes created on frequently queried fields (userId, projectId, status, projectType)

12. Environment Configuration

12.1 Required Environment Variables

- ✓ DATABASE_URL — PostgreSQL connection string
- ✓ JWT_SECRET / JWT_REFRESH_SECRET — 256-bit secrets for signing tokens
- ✓ JWT_ACCESS_EXPIRES_IN (default 15m) / JWT_REFRESH_EXPIRES_IN (default 7d)
- ✓ CF_ACCOUNT_ID, CF_R2_ACCESS_KEY_ID, CF_R2_SECRET_ACCESS_KEY, CF_R2_BUCKET — Cloudflare R2 storage (or 'demo')
- ✓ RESEND_API_KEY / RESEND_FROM_EMAIL — Email service configuration
- ✓ APP_BASE_URL / API_BASE_URL — Frontend and backend URLs for email links
- ✓ ADMIN_EMAIL — Admin email for receiving notifications
- ✓ COMPANY_NAME — Company name for emails
- ✓ NODE_ENV — development | production

12.2 Database Initialization

- ✓ pnpm db:migrate — Run all migrations
- ✓ pnpm db:seed — Seed initial data (optional)
- ✓ pnpm prisma:generate — Regenerate Prisma client

13. Deployment & Build

13.1 Build System

- ✓ pnpm build:api — Compiles TypeScript to dist/
- ✓ pnpm start:prod — Runs compiled application
- ✓ pnpm dev:api — Runs with hot-reload

13.2 Package Management

- ✓ pnpm monorepo workspace with apps/api and apps/web
- ✓ Lock files for reproducible builds
- ✓ All dependencies use exact versions (no ranges)

13.3 Linting & Formatting

- ✓ ESLint — Code linting with rules configured
- ✓ Prettier — Code formatting
- ✓ pnpm lint — Auto-fixes issues; pnpm format — Applies prettier

13.4 Testing

- ✓ Jest unit test framework configured
- ✓ Test suite with services spec files present
- ✓ pnpm test — Runs all tests

14. Final Verification Checklist for Boss

14.1 Core Features Verified

- ✓ **User Management** — 4 roles, authentication, password reset, account lifecycle
- ✓ **Project Lifecycle** — Create, assign, work, deliver, send to client; full status history
- ✓ **RFI System** — Engineers raise, admins answer or forward to client
- ✓ **File Management** — Upload intake files, download deliverables; S3 + demo mode
- ✓ **Client Delivery** — Flexible delivery (attachment or links), invoice generation, merchant fee tracking
- ✓ **Email Notifications** — 10+ automated email triggers with personalization
- ✓ **Financial Tracking** — Pricing, merchant fees, invoice generation, weekly reports with KPIs
- ✓ **Audit & Compliance** — 18+ audit event types, complete access logs, role-based filtering
- ✓ **Security** — JWT auth, password hashing, rate limiting, account lockout, RBAC, input validation
- ✓ **Database** — PostgreSQL schema with 11 tables, proper foreign keys, indexes

14.2 Quality Assurance

- ✓ No hardcoded credentials — all sensitive data in environment variables
- ✓ Proper HTTP status codes, meaningful error messages (no data leaks)
- ✓ All list endpoints support pagination
- ✓ Input validation on all endpoints
- ✓ Engineers cannot see client details or pricing via any endpoint
- ✓ All sensitive actions logged with actor and timestamp
- ✓ Graceful error handling; failures logged but don't crash system
- ✓ Secure presigned URLs; no public file access

14.3 Operational Readiness

- ✓ .env example file provided; all required variables documented
- ✓ Version-controlled migrations; easy to deploy
- ✓ Dev and production start scripts available
- ✓ Errors logged; can be configured per environment
- ✓ This checklist + README + schema comments provided

Pre-Launch Checklist

- Database credentials configured for production PostgreSQL
- AWS S3 / Cloudflare R2 bucket created and credentials set
- Resend / SendGrid / SMTP provider configured (or use demo mode)
- SSL/TLS certificate installed on production server
- Environment variables (.env) set in production
- Database migrations run (pnpm db:migrate)

- Application built (pnpm build:api)
- Production server started (pnpm start:prod)
- Health check endpoint tested
- Email notifications tested end-to-end
- Audit logs verified

Support & Maintenance

- ✓ **Logs Location** — Application logs available via console/Docker logs
- ✓ **Database Backups** — Regular PostgreSQL backups recommended
- ✓ **Security Updates** — Monitor dependencies for security patches
- ✓ **Monitoring** — Recommend an APM tool (e.g., New Relic, DataDog) for production

Conclusion

The ACE Services Portal is a complete, production-ready application with all core features implemented: multi-role project management, secure authentication & authorization, automated email notifications with tracking, financial management & reporting, comprehensive audit logging, flexible file handling (S3 + demo mode), an RFI workflow for client communication, role-based view filtering for data privacy, and professional invoice generation.

Status: Ready for Deployment

Prepared by Development Team | August 25, 2026 | Version 1.0 Final